Evidence-order and contamination sensitivity analysis

Two-indicator, full-evidence-bank experimental design

Aim and analytical scope

A focused sensitivity analysis examined whether the frozen Stage 2 assessment was affected by (1) the order in which otherwise identical evidence records were presented and (2) the addition of deliberately low-quality synthetic records. Two indicators were purposively selected from substantively different DEMA subdimensions: A.1.2, concerning the identification and role of the coordinator within the A.1 Roles subdimension, and B.1.3e, concerning data validity within the B.1 Data supply subdimension. The two-indicator design provided contrasting institutional and technical assessment tasks while remaining compatible with the model's output-token limit.

The analysis was descriptive and evaluated pipeline robustness rather than country-level maturity. The unit of analysis for the order experiment was an indicator assessment under one evidence ordering. The unit of analysis for contamination was the matched clean–contaminated indicator pair. Results were therefore interpreted as two indicator-level case studies and not generalized statistically to all 31 DEMA indicators.

Experimental conditions

Condition	Calls	Evidence bank	Purpose
Clean reorderings	5	Same 622 genuine records; five seeded orders	Estimate order sensitivity
Contaminated	1	Clean ordering 1 + 155 synthetic controls	Estimate one matched contamination contrast

For the order analysis, the same 622 genuine evidence records and stable neutral EV-#### identifiers were retained in every call. Only list order changed. Five reproducible seeds (20261101–20261105) generated five permutations. Both indicators were scored together in each call, ensuring that they encountered the same evidence ordering.

The contamination condition was paired with clean ordering 1 (seed 20261101). The relative order of its 622 genuine records was preserved and 155 controls were inserted at seeded positions. The treatment contained five controls for every DEMA indicator. Thus, five controls targeted A.1.2, five targeted B.1.3e and 145 represented broader evidence-bank noise. Because contamination was executed once, it provides a matched case contrast rather than a replicated estimate of the average contamination effect.

Frozen scoring procedure

All six primary calls used the frozen Stage 2 P5 Langfuse prompt (version 1; recorded SHA-256 beginning A3734A364931), GPT-4.1, temperature 0, an unchanged output schema and a 32,768-token response allowance. Prompt identity was hash-verified before execution. The country, indicator definitions, evidence content and validation procedure were held constant. Calls were traced in Langfuse and flushed after completion. Analyst-only mappings retained record type and original identity; these labels were not supplied to the scoring model.

Outcome measures

Metric	Definition and interpretation
Phase distribution	Number of the five clean reorderings assigned to each phase.
Pairwise phase agreement	Proportion of the 10 possible clean-run pairs with the same phase; 100% indicates order-invariant scoring.
Supporting-set Jaccard	$\|S_1 \cap S_2\| / \|S_1 \cup S_2\|$ for positively cited evidence sets; 1 indicates identical selection.
Exact supporting-set agreement	Proportion of clean-run pairs with exactly the same positive evidence-ID set.
Limiting-set Jaccard	Jaccard similarity between contrary-or-limiting evidence sets across reorderings.
Paired phase change	Contaminated phase minus the phase from matched clean ordering 1.
Genuine-support retention	|Sclean ∩ Streatment| / |Sclean|; proportion of matched-clean support retained after contamination.
Control role	Controls in evidence_supporting_score were positive contamination; controls occurring only in contrary_or_limiting_evidence were limiting references.

Score stability and evidence-selection stability were treated as distinct outcomes. An unchanged phase did not imply an unchanged justification. Likewise, use of a synthetic record as limiting evidence was not classified as positive contamination because it qualified or rejected an evidentiary claim rather than supporting maturity.

Construction of synthetic controls

The synthetic controls instantiated five recurrent forms of maturity-inadequate material: an agenda mention, presentation attendance, a generic external comment, a future possibility and a glossary reference. They were introduced solely for the experiment and did not originate from the documentary corpus. Each record used topical indicator language but explicitly lacked evidence of implementation, institutionalization, coverage or outcomes.

Indicator	Control type	Evidence statement supplied
B.1.3e	Future Possibility	Participants expressed possible future interest in Data quality: validity.
A.1.2	Agenda Mention	A meeting agenda in Togo mentioned Roles are known & identified, for the coordinator as a discussion topic.
B.1.3e	Agenda Mention	A meeting agenda in Togo mentioned Data quality: validity as a discussion topic.
A.1.2	Future Possibility	Participants expressed possible future interest in Roles are known & identified, for the coordinator.

Validation and protocol deviations

All six calls returned both requested indicators, valid JSON and valid evidence identifiers; no citation repairs were required. Three B.1.3e responses contained 157–158 words rather than the prompt-specified minimum of 180 words (clean reorderings 2 and 3 and the contaminated call). These outputs were retained because the deviation concerned explanatory length rather than phase completeness, schema validity or citation integrity, and it is reported as a protocol deviation when interpreting the results.